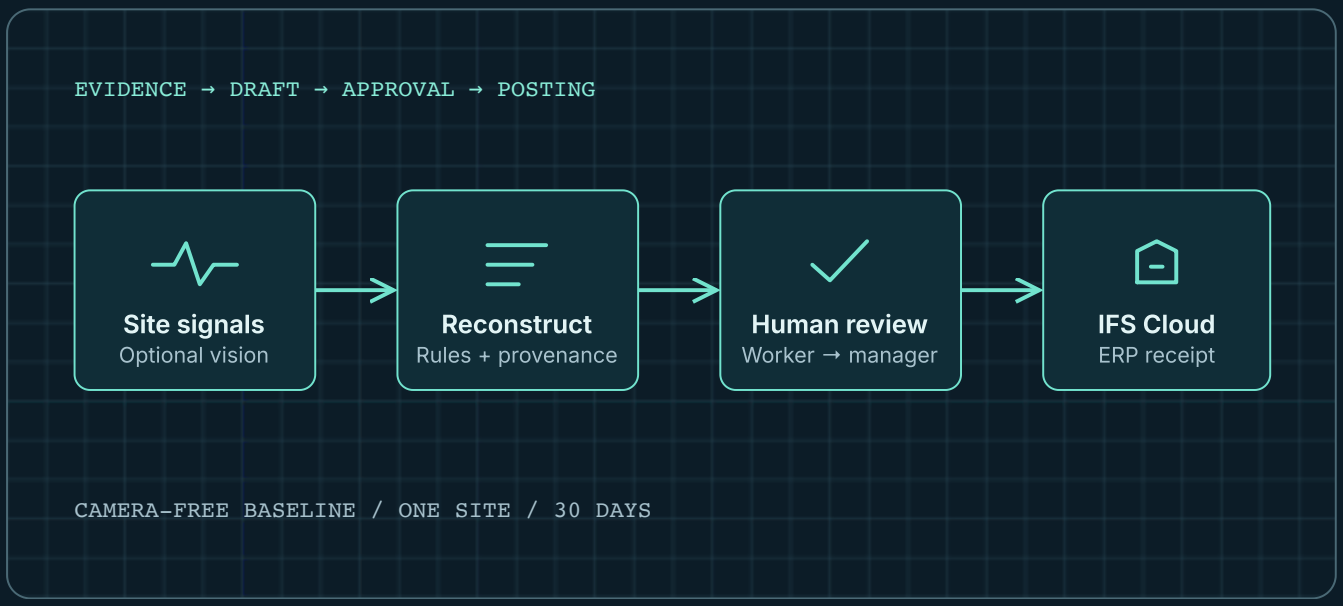
Zero-Touch Week.

From operational signals to accurate timesheets.

An edge-first design for IFS Cloud field teams. Reconstruct a draft from operational evidence, surface uncertainty and give people control over what gets posted.



Technical architecture brief · v1.0

Open the interactive showcase

BUSINESS CONTEXT

Build the draft while the work happens.

Delayed time entry creates corrections, missing expenses, late reimbursements and unbilled work. Start with one site and the evidence already available to field workers.

Time tolerance

Approximately 30 minutes. Confirm whether the tolerance applies to individual boundaries, total daily time, or both. Measure both during shadow mode.

Activity codes

Travel, work, waiting and break are illustrative. Each tenant maps permitted activity and pay codes under a versioned policy.

Three people-facing responsibilities

Role	Responsibility	Visible outcome
Worker	Review proposed intervals and fix missing or incorrect evidence.	A confirmed, editable draft with its sources.
Manager	Resolve exceptions and authorize a specific revision.	Approval, reasoned rejection or request for changes.
ERP owner	Validate tenant mapping and reconcile accepted writes.	An authoritative IFS transaction or a visible failure.

Evidence boundaries

An assigned six-hour order and seven hours of site presence support a possible allocation. Work completion comes from a validated status event, operational record or human confirmation. Camera observations represent crossings of people in a zone. Named identity requires a separately authorized check-in or badge source.

> **Product principle.** Automate data capture, not accountability. Missing evidence stays visible. A policy engine flags possible breaks or overtime and leaves any correction to an authorized person.

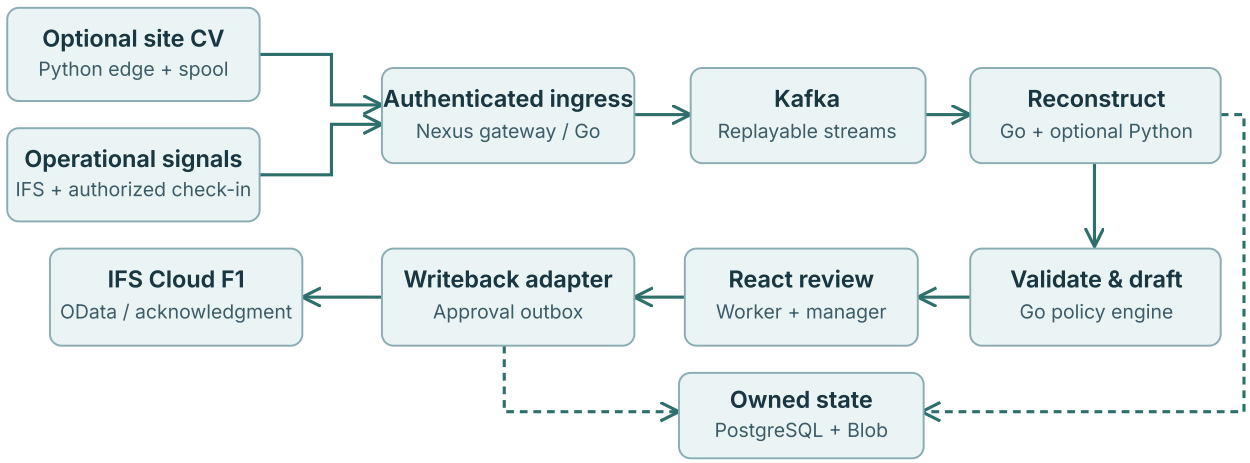
First scope

Time entry for one worksite. Expense reimbursement remains a later IFS workflow that would use receipts and worker attestations. The system does not invent hours, mileage, breaks, overtime or completion from incomplete evidence.

REFERENCE ARCHITECTURE

Separate evidence, decisions and posting.

Kairos/Nexus owns draft revisions and processing state. IFS Cloud F1 owns the authoritative approved time and expense transactions.



Solid arrows show the decision flow. Dashed lines link components to application-owned state. Confirm platform boundaries and gateway capabilities with the IFS owners.

One-site pilot

Single Go deployment and a PostgreSQL outbox, a React review UI, authorized check-ins and a supported IFS test feed. Run in shadow mode. Optional vision sits on a separate edge box after approval.

Regional fleet

Partition Kafka streams and scale Go consumers independently on Azure. Use AKS when operating needs justify it. Run Python inference in isolated workloads with separate capacity and model release controls.

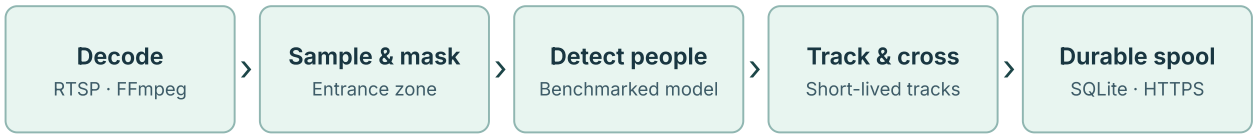
Observe what the worker sees

Use OpenTelemetry across ingestion, replay, review and writeback. Export bounded-cardinality metrics to Prometheus/Grafana or Azure Monitor. Alert on stale evidence, consumer lag, spool age, draft delay and unacknowledged ERP transactions. Keep personal fields out of logs and labels.

EDGE VISION

Keep video near the gate.

Enable only at approved entrance and exit zones. Run a camera-free baseline first and measure incremental value before retaining this capability.



Stream decoding and sampling are distinct costs. Measure throughput across the full pipeline.

Crossing semantics

Track an object for seconds within the approved field of view. Require stable observations on opposite sides of a virtual line with hysteresis. Deduplicate oscillations. Emit zone, direction, capture time and quality flags. Use a short-lived track token only inside the gateway.

Offline delivery

Use durable SQLite with bounded disk usage, a stable event ID and exponential backoff. Delete only after durable ingestion acknowledgment. Monitor clock offset, queue age and missing intervals. If the quota is exhausted, record a coverage incident and use the check-in path.

Benchmark and license gate

Compare CPU and accelerators using representative resolution, lighting, crowding, FPS, decode, detection and tracking. Measure missed/false entries and exits against labeled crossings. Review Ultralytics code and weights for AGPL-3.0 obligations and Enterprise licensing before proprietary use. The repository contains neither package nor weights.

> **Identity boundary.** No face, gait or helmet identity, persistent re-identification, activity scoring or disciplinary decisions. A crossing can corroborate site context only.

Reconcile across the ERP boundary.

Event delivery, approval and ERP posting are separate durable states. Each has a distinct recovery path.


```
{
  "schemaVersion": 1, "eventId": "evt-cross-0101",
  "tenantId": "demo-tenant", "siteId": "gate-a",
  "source": "edge.entrance", "type": "presence.crossing.observed",
  "occurredAt": "2026-11-09T07:46:02Z",
  "processedAt": "2026-11-09T07:46:05Z",
  "correlationId": "trace-demo-0123", "workOrderId": null,
  "data": {"direction": "in", "quality": "two-sided-track"}
}
```


Kafka, inbox and outbox

Partition camera streams by tenant/site and worker-authorized streams by tenant/worker. Per-key ordering does not imply event-time order across feeds. Commit application state, deduplication key and outbox records in one PostgreSQL transaction, then commit the consumer offset. Quarantine poison records with a reason and an audited replay path.

IFS Cloud F1 contract discovery

Discover supported OData projections and permitted actions in the target release's API Explorer. Confirm legal entity, worker/project codes, authorization, transaction semantics, external reference, status query and versioning with the IFS owners. The Oracle and PL/SQL business boundary remains inside F1. Ingest through a documented event mechanism and reuse approved Nexus gateways where available.

State	Condition and recovery
Approved	Lock the revision and atomically enqueue a writeback command with a stable tenant/revision key.
Posting	Lease command. Retry transient failures within bounds. After an ambiguous response, resolve by external reference before sending again.
Posted	Persist the authoritative IFS transaction ID and acknowledgment. Reconcile against IFS records.
Needs reconciliation	Surface unknown outcomes and contract/validation failures. If reliable lookup is unavailable, require operator resolution.

RECONSTRUCTION AND EVALUATION

Expose uncertainty before approval.

Start with deterministic state transitions and event lineage. Model outputs can rank suggestions after evaluation.

Evidence case	Expected behavior
Out of order or duplicate	Deduplicate by tenant/source/event ID. Order by event time plus a deterministic sequence. Recompute within the lateness window.
Missing end, break or overlap	Keep the interval open or contested. Surface the policy exception and let the worker correct it.
Skew, DST and overnight	Store UTC instants, site IANA zone and device offset estimate. Split approved totals according to tenant-local shift policy.
Several work orders	Keep candidate assignments and provenance. Require clarification where signals conflict. No double-counted payable minutes.
Retrospective status	Rebuild affected unapproved revisions. Route approved items through a superseding correction workflow.

Candidate models

Benchmark a state machine first. Compare interpretable logistic regression with calibrated gradient-boosted trees or random forest for nonlinear interactions. Only consider LLM extraction for permitted ambiguous text, with quoted source spans, schema validation and no direct policy or posting authority.

Use assignment, timing overlap, event sequence, planned duration, role/task type, prior approved allocations and authorized attendance where justified. Avoid unnecessary sensitive features. Freeze feature availability at draft time.

> **Evaluation.** Split forward in time and hold out entire tenants/sites. Exclude corrected totals and other post-decision fields from features. Compare coverage and error at calibrated thresholds, Brier score, reliability curves and harmful error by site, role and evidence availability.

Use a validation set to tune abstention against the cost of wrong hours and allocations. Monitor drift, keep shadow evaluation and fall back to deterministic rules. A score is an estimate of evidence quality or likelihood, with no approval authority.

Size from measured throughput.

Treat footprint, frame size, utilization, network and prices as scenario inputs. Establish decode-inclusive throughput before selecting hardware.

```
frames/day = cameras × sampled_fps × active_hours × 3600
raw GB/day = frames/day × sampled_frame_kB / 1,000,000
edge workers/site = ceil(site_fps / (worker_fps × utilization))
cloud workers = ceil(frames/day /
                (worker_fps × useful_seconds × utilization))
```

Example input	Illustrative result	Implication
2 cameras, 2 fps, 10 h/day	144,000 sampled frames/day	4 fps/site while active.
90 kB/frame	12.96 GB/day · 2.88 Mbps live	Budget protocol overhead and backlog drain.
12 fps edge CPU, 70% target	1 worker/site	8.4 fps usable under the scenario.
120 fps cloud GPU, 3 h window, 10 min startup	1 worker · ~29 min inference	Upload and cold start add elapsed time.

Compute and queue strategy

Benchmark edge CPU, local GPU and cloud CPU first. If permitted centralized batches need acceleration, request GPUs in a versioned Python Kubernetes Job. Use GPU affinity, taints/tolerations and driver validation. Drive job count from backlog using KEDA ScaledJob where appropriate and evaluate a dedicated autoscaled AKS node pool. Exercise quota exhaustion, retries, dead letters, scale-to-zero and cold-start latency.

Validate candidate Azure families against region, quota, pricing, GPU memory and measured performance. Include transfer, amortization, billed idle time and retained evidence. No quoted hardware price or benchmark is used here.

> **Sampling.** With periodic frames and random phase, `min(1, fps × seconds visible)` estimates the chance of one sample. Tracking often needs several frames. Measure precision and recall for entry and exit separately.

Give every piece of evidence a purpose.

Treat camera use as a separate decision for each customer, purpose and jurisdiction. Require a camera-free alternative.

Boundary	Control	Verification
Gateway	Scoped identity, outbound TLS, secret rotation, signed images, restricted camera VLAN.	Revoke a device and attempt cross-site submission.
Application	Tenant and assignment claims, least-privilege worker/manager roles, authorized actions by revision.	Exercise cross-tenant IDs, object access and replay.
Data	Tenant-keyed PostgreSQL, scoped Blob access, encryption, managed keys where required, region-aligned backups.	Restore into isolation and confirm retention/deletion across versions, snapshots and soft deletes.
Audit	Provenance, policy/model version, reviewer decision, correction and ERP acknowledgment.	Reconstruct decisions without logging raw images or employee data in telemetry.

Camera activation criteria

Document purpose, compatibility of any reuse, proportionality and lawful basis. Employee consent may be unsuitable given the power imbalance. Review applicable labor law and employee information or consultation obligations. Complete a DPIA where required and resolve outstanding risks before activation. In France, review CNIL guidance with the controller and counsel.

Use scoped badge or explicit check-in data only where authorized. Keep employee references out of camera events. Even anonymous observations can become personal data when combined with other records, so govern that join and restrict it to the minimum needed.

> **Data minimization.** Prefer crossing metadata with short retention. Approved evidence objects have a defined expiry and access audit. No biometric identification, continuous productivity scoring, automatic disciplinary use or unexplained payroll deductions.

ONE SITE · 30 DAYS

Earn the right to expand.

Use one customer and one site. Compare the workflow to the current process and camera-assisted reconstruction to the camera-free baseline where permitted.

Days	Delivery and test	Decision gate
01–07	Discover IFS contracts, confirm scope, capture manual baseline, review permissions and profile hardware.	Approve cohort, metrics, tolerance, lawful basis and evidence handling.
08–14	Build and replay the operational-event pipeline. Exercise offline buffering, duplicates, gaps and clock handling.	Deterministic replay and visible failure states. Vision remains conditional.
15–21	Shadow drafts against approved records. Run worker correction and manager review. Measure errors by shift and evidence coverage.	Production payroll stays disconnected. Review high-impact false approvals.
22–30	Run assisted review, test ERP acknowledgment in an approved environment and measure cost and time saved.	Approve, extend or stop based on signed thresholds and resolved privacy risks.

Scorecard

Track median/p90 absolute error and the share of daily totals and boundaries within tolerance. Measure per-category precision/recall, allocation match, worker corrections, worker/manager minutes, missing signals, false and missed crossings, harmful approvals, cost per site/day and compliance gates.

Agree denominators and ground-truth labeling before shadow mode. Treat missing records as a separately reported coverage outcome. Include cameras-off, connectivity-limited and atypical shifts in the sample. Compare results by cohort with uncertainty intervals.

> **Illustrative hypotheses only.** At least 90% of eligible daily totals within the agreed tolerance, measurable worker time saved and zero observed harmful approvals. Confirm sample size and acceptance criteria with the tenant before use.

RISKS AND NEXT STEPS

Review the unresolved decisions.

Progress through gated releases. Document the observed errors and cost before committing to a regional fleet.

Risk	Response
Vision adds little value	Keep the camera-free workflow. Add approved entrance vision only for measured improvement that justifies its burden.
Ambiguous payroll precision	Agree acceptable error for boundaries and totals and what constitutes compensable work. Version tenant-specific rules.
ERP outcome is uncertain	Confirm the projection and reconciliation mechanism in the target release. Block unsafe retries when outcome lookup is unavailable.
Tenant or regional drift	Validate model behavior and worker rights per jurisdiction. Roll out by cohort and monitor changes in evidence coverage.
Scaling creates cost or privacy exposure	Prefer metadata, measure demand and per-site load, and validate quota, residency, restore and deletion before expansion.

Before the first customer pilot

Confirm the IFS integration owner, worker evidence and correction paths, one-site pilot sponsor, policy ownership, legal review and comparison dataset. Identify who can disable vision and pause writeback.

Primary references

IFS Cloud · API discovery and projections

IFS Cloud · API usage policy

Apache Kafka · Delivery semantics

Microsoft · AKS GPU scheduling

Microsoft · AKS scaling

Microsoft · Blob lifecycle deletion

EUR-Lex · GDPR Articles 5, 6 and 35

CNIL · Workplace video monitoring

Ultralytics · AGPL and Enterprise licensing

Zero-Touch Week · Source and technical notes